# Screening Colonoscopy for Colorectal Cancer in an Average-risk Population: Systematic Review and Meta-analysis

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**ABSTRACT** Colorectal cancer (CRC) is a major health concern, ranking as the third most common cancer in the United States. Screening programs, especially colonoscopy, play a crucial role in preventing CRC by removing and detecting polyps or early-stage cancers. Despite inherent risks, colonoscopy's effectiveness in saving lives is significant. In this review, we analyzed the outcomes of screening colonoscopies in an asymptomatic population for 15 years, focusing on detection rates and complications. We compared the data with previous meta-analyzes to evaluate changes in efficacy and safety. We conducted a systematic search of medical literature databases (1 January 2012 to 31 December 2023) for English-language studies on CRC screening colonoscopy. Our inclusion criteria comprised complete articles with over 500 participants with extractable data and a focus on screening colonoscopy outcomes in average-risk populations. In total, 2,897,025 people were screened, most (99.6%) were asymptomatic and were an average-risk population. Colonoscopy was complete and reached the cecum in 97–99% of the procedures. CRC was found in 0.5% (95% confidence interval [95%CI] 0.4–0.7%) of the participants. Advanced adenoma was found in 7.6% (95%CI 6.2–9.3%) of the cases. Complications were rare. Perforation developed in 0.022% of the cases and bleeding in 0.148%. Our findings exhibited a significant increase in the detection yield of adenomas and advanced adenomas with low complication rates, which shows that colonoscopy is feasible and suitable for screening for CRC in asymptomatic people.

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**KEY WORDS:** adenoma detection rate, advanced adenoma, colorectal cancer (CRC), perforation, serrated adenoma

Colorectal cancer (CRC) presents a significant public health challenge. It is the third most common cancer in the United States, ranking second only to lung cancer in cancer-related mortality, thus it is an attractive target for population screening [1]. As of 2021, nearly 500,000 new CRC cases were reported, with over 52,980 attributed deaths. Over the past three decades, a notable decrease in CRC incidence rates has been observed, particularly in individuals aged 50 years and older. This decrease has been attributed to increased awareness and adherence to screening practices. This decline in incidence has been accompanied by a reduction in mortality rates by more than 30%, primarily stemming from advances in early detection and therapeutic interventions [2].

A significant majority of CRC cases develop from adenomatous polyps, commonly detected in nearly 50% of the elderly population. However, only a small fraction of the polyps undergo malignant transformation. The asymptomatic nature of most polyps contributes to the evasive clinical detection during the early stages of the disease. Colorectal cancer screening programs are founded on the rationale that the removal of adenomatous polyps will prevent CRC development, while early detection of localized, superficial cancers in asymptomatic individuals will enhance the surgical cure rate. Colonoscopy, fecal occult blood testing (FOBT), stool DNA, digital rectal examination, rigid and flexible sigmoidoscopy, computed tomography colonography (virtual colonoscopy), and liquid biopsy have been considered for CRC screening. All these procedures have benefits and disadvantages.

Colonoscopy shows the highest sensitivity rates and is considered the gold standard for detecting and preventing CRC in many countries [3-5]. The effectiveness of colonoscopy in detecting and removing polyps has played a crucial role in reducing both CRC incidence and mortality rates [6]. In recent years, new advancements have been tested and implemented to enhance the detection rates of CRC precursors, such as adenomatous and serrated polyps [7]. Such advancements include the utilization of trained artificial intelligence (AI), as well as the use of a cap (a transparent hood or cap attachable to the tip of the colonoscope) that improves visualization [8]. Yet, colonoscopy carries some inherent risks [9]. Perforation, although rare, can

occur, especially in cases with underlying conditions that make the colon wall more fragile. Polyp removal during colonoscopy may cause minor bleeding, which usually stops on its own, but in rare cases may necessitate further intervention. Sedation carries some risks, such as allergic reactions or adverse effects of the respiratory and cardiovascular systems. A meta-analysis conducted 15 years ago showed an adenoma detection rate (ADR) of 19% and an advanced adenoma detection rate (AADR) of 5%, as well as a CRC detection rate of 0.78% [10].

**COLONOSCOPY SCREENING FOR ASYMPTOMATIC INDIVIDUALS DEMONSTRATES ENHANCED EFFICACY IN BOTH DETECTING AND PREVENTING COLORECTAL CANCER.**

In this article, we analyzed the outcomes of screening colonoscopy in an asymptomatic population over the past 15 years. We conducted a systematic review and meta-analysis on screening colonoscopy.

## METHODS

### IDENTIFICATION OF STUDIES AND DATA EXTRACTION

We searched PubMed, EMBASE, Scopus, and CENTRAL databases from 1 January 2012 until 31 December 2023 to identify human studies written in English. The following search text and/or medical topic heading (MeSH) terms were used: *colorectal cancer* OR *CRC* [All Fields] AND *screening colonoscopy* [MeSH Terms]. To retrieve original studies, we also conducted a manual search of all review articles and editorials. Hand searches included article bibliographies. This meta-analysis was performed according to the PRISMA extension statement for interventions [11].

### SELECTION CRITERIA: PRIMARY ENDPOINTS

Inclusion and exclusion criteria were decided before starting the study investigation. Articles that were written in English, included data that could be extracted, and focused on the outcomes of screening colonoscopy projects in average-risk populations were included. We only selected studies that separated the results of finding CRC, adenoma, advanced adenoma, and serrated adenoma.

### HETEROGENEITY, SENSITIVITY, AND PUBLICATION BIAS

The heterogeneity of the studies was calculated using the Cochran Q test and $I^2$ inconsistency index, which was considered to be present if the Q-test *P*-value < 0.10. The higher the $I^2$, the greater the heterogeneity [12]. The sensitivity testing was conducted by removing individual studies from the overall result, calculating papers with high and low ADR separately, and calculating ADR excluding the study with a higher event rate (ER). The publication bias was analyzed using a funnel plot complemented by Begg-Mazumdar and Egger statistics [13]. We also used the Slim methodological index for non-randomized studies (MINORS) for the evaluation of non-randomized study quality for meta-analysis [14].

### STATISTICAL ANALYSIS

We used Comprehensive Meta-analysis Software (Version 4, Biostat Inc., Englewood, NJ, United States). Pooled ERs and 95% confidence intervals (95%CIs) were calculated to compare colonoscopy results in individual studies by using the random effects model.

## RESULTS

### A SYSTEMATIC REVIEW OF THE SELECTED STUDIES

***POX et al. (Germany, 2012) [15]***

The authors presented the findings from screening colonoscopies conducted within the first 6 years of the German screening program, which spanned from January 2003 to December 2008. Data were gathered from over 2100 practices, totaling 2,821,392 screening colonoscopies. The study focused on asymptomatic, average-risk individuals aged 55–74 years. Results presented an ADR of 19.4%, showing a higher prevalence in men. Advanced adenomas were found in 6.4% of patients, while CRC was identified in 25,893 subjects (0.9%).

***Jover et al. (Spain, 2013) [16]***

The authors examined modifiable factors directly associated with the endoscopic procedure that influenced the ADR in screening colonoscopies. The participants included 4539 asymptomatic individuals aged 50 to 69 years undergoing direct screening colonoscopy. The findings revealed that the factors independently related to the ADR were a mean withdrawal time longer than 8 minutes in normal colonoscopies and a preparation procedure spanning 2 days. For advanced adenomas, only withdrawal time maintained statistical significance in the multivariate analysis. Cecal intubation was achieved in 85% of all colonoscopies, ADR was 28.6% and advanced ADR was 11.4%.

***Butterly et al. (USA, 2014) [17]***

Using data from the New Hampshire Colonoscopy Registry, the authors explored how an endoscopist's with-

drawal time during normal colonoscopies impacted adenoma, sessile serrated polyps, and hyperplastic polyp detection. Among 7972 patients from 2009 to 2011, 5521 were classified as average risk. ADR and sessile serrated adenoma detection rate (SSADR) were calculated based on the median endoscopist withdrawal time during normal examinations. The authors found that the ADR was highest among endoscopists with a median withdrawal time of 9 minutes, while the detection of sessile serrated polyps and hyperplastic polyps peaked at 8–9 minutes. ADR reached 33.6% and SSADR 8%.

***Ross et al. (USA, 2015) [18]***

The authors assessed the ADR and SSADR in screening colonoscopies before and after the implementation of quality improvement measures. The study involved 2833 average-risk CRC screening patients aged 50 to 75 years. ADR/SSADR of the physicians were documented, and prospective data collection was initiated to record their ADR and SSADR after informing them of their baseline rates. The study showed that exceeding current ADR standards is achievable, but despite improvement over time, the changes did not reach statistical significance. ADR was 41%, SSADR was 8.2%, and advanced adenoma detection rate (AADR) was 8.6%.

***Bretthauer et al. (Norway, 2016) [19]***

The authors explored the participation rate, ADR, performance, and adverse events associated with population-based colonoscopy screening across several European countries. This randomized clinical trial included 94,959 average-risk patients aged 55 to 64 years in Poland, Norway, the Netherlands, and Sweden from 2009 to 2014. The participants were divided into two arms: colonoscopy screening (n=12,574) or no screening (n=82,385). One perforation (0.01%) and 18 cases of bleeding after polypectomy (0.14%) were observed. Results showed ADR and CRC diagnosis rates of 30.7%, and 0.5%, respectively.

***Zwink et al. (Germany, 2017) [20***

The authors examined the occurrence of complications of colonoscopies conducted as part of the German screening program. It included 5252 patients without a history of CRC or prior polypectomy who underwent screening colonoscopies between 2010 and 2013. The study utilized questionnaires and verified self-reported complications by reviewing colonoscopy records. Among the participants, 16 reported physician-confirmed bleeding (0.30%) and 4 experienced physician-confirmed perforation (0.08%) during or within 4 weeks of the procedure. The ADR and AADR were 11% and 1%, respectively, and the CRC detection rate was 1%.

***Wong et al. (Hong Kong, 2019) [21]***

The authors compared the ADR, AADR, and the number of adenomas per colonoscopy (APC) in direct screening colonoscopy (DSC) versus FIT-positive colonoscopy (FITC). Asymptomatic individuals over 50 years of age, with complete colonoscopies to the cecum and at least fair bowel preparation were eligible for inclusion in the study. Overall ADR, sex-specific ADR, overall AADR, and overall APC were compared between the two screening methods. The results showed that overall ADR, male-specific ADR, female-specific ADR, and overall AADR in FITC were significantly higher than the corresponding values for DSC. ADR in the DSC arm was 37.5%, advanced ADR 4.9% and CRC was diagnosed in 0.1%.

***Kim et al. (Korea, 2020) [22]***

The authors assessed the efficacy of cap-assisted chromoendoscopy (CAP/CHROMO) compared to standard colonoscopy (SC) in detecting adenomas. In this trial, asymptomatic subjects aged 45–75 years undergoing their first colonoscopy were randomly assigned to either the CAP/CHROMO group or the SC group. The primary outcome focused on the ADR. Among the 1,905 subjects enrolled, the CAP/CHROMO group demonstrated a significantly higher ADR (54.4%) compared to the SC group (44.9%). Moreover, in the female subgroup, CAP/CHROMO showed a significantly higher detection rate of proximal serrated polyps than SC. However, there was no significant difference in AADR between the two groups.

***Ladabaum et al. (USA, 2021) [23]***

The study introduced the Adenoma Detection Rate–Extended to all Screening/Surveillance (ADR-ESS) score, an automated colonoscopy audit program based on standardized clinical documentation. In total, 15,253 colonoscopies were performed, ADR and SSADR were calculated and the correlations between ADR-ESS, and ADR were investigated. Patients were categorized into groups: screening first and screening not first (considered average risk). Individuals with a family history of CRC or a positive screening at follow-up were classified as high risk and were excluded from the study. The results showed that ADR-ESS increased the audit volume by 3.4 times, providing more precise and stable outcomes than ADR. There were correlations between ADR and SSADR. ADR reached 36.3%, SSADR 7.2%, and AADR 6.7%.

*Xu et al. (China, 2023) [24]*

The authors compared artificial intelligence (AI)-assisted colonoscopy with conventional colonoscopy for ADR in an asymptomatic population. Asymptomatic patients aged 45–75 years undergoing CRC screening by direct colonoscopy were recruited. The primary outcome was the overall ADR. Secondary outcomes included the mean number of adenomas per colonoscopy, ADR based on the endoscopist's experience, and colonoscopy withdrawal time. A total of 3,059 subjects were randomized to either AI-assisted colonoscopy or conventional colonoscopy. ADR and AADR rates were significantly higher in the AI-assisted colonoscopy group. In the AI-assisted colonoscopy group ADR, AADR, SSADR, and CRC were 39.7%, 6.7%, 1.3%, and 0.2%, respectively, in comparison with 33.5%, 5.1%, 1.5%, and 0.6% in the conventional colonoscopy group. The study described two informative groups based on two different studies.

**ALTHOUGH COLONOSCOPY HAS SIGNIFICANTLY IMPROVED IN EFFICACY, THERE MAY BE PROCEDURAL COMPLICATIONS**

## META-ANALYSIS RESULTS

We found 10 studies, with 12 sub-studies, published between 1 January 2012 and 31 December 2023, of screening colonoscopy for CRC detection in average-risk, asymptomatic populations, and met our inclusion criteria [15-24] [Figure 1]. In total, 2,897,025 people were screened, most of whom were (99.6%) asymptomatic and were part of an average-risk population. They came from 6 countries (United States, Germany, Spain, Norway, China, and South Korea) in 12 years [Table 1]. There was a similar number of men and women, with an average age of 58 years. The number of patients per study ranged from 935 (913 average risk) to 2,821,392 (all of them average risk). The second biggest study contained 3142 patients, all of them of an average risk for CRC. Kim et al. [22] compared cap-assisted chromo-colonoscopy with standard colonoscopy and found ADR of 54.4% and 44.9%, respectively. This study dealt with two different articles. Xu et al. [24] compared AI-assisted colonoscopy with conventional colonoscopy in 3059 patients. ADR was 39.7% and 33.5%, respectively. This study dealt with two different articles. The largest study by Pox and colleagues [15] described the first 6 years of the German screening program. This prospective cross-sectional study included 2,821,392 screening colonoscopies; thus, the meta-analysis was profoundly influenced by its results. To overcome this potential bias, we calculated the data with and without this study. Only 2499 of 2,897,025 (0.086%) of the patients were a high-risk population. The results of the meta-analysis of diagnostic yields for ADR, AADR, SSADR, and CRC are presented in Table 2. ER for ADR, AADR, SSADR, and CRC were 0.353 95%CI 0.289–0.423, 0.076 95%CI 0.062–0.093, 0.023 95%CI 0.014–0.038, and 0.005 95%CI 0.004–0.007, respectively.

**Figure 1.** Flowchart of studies included in the meta-analysis

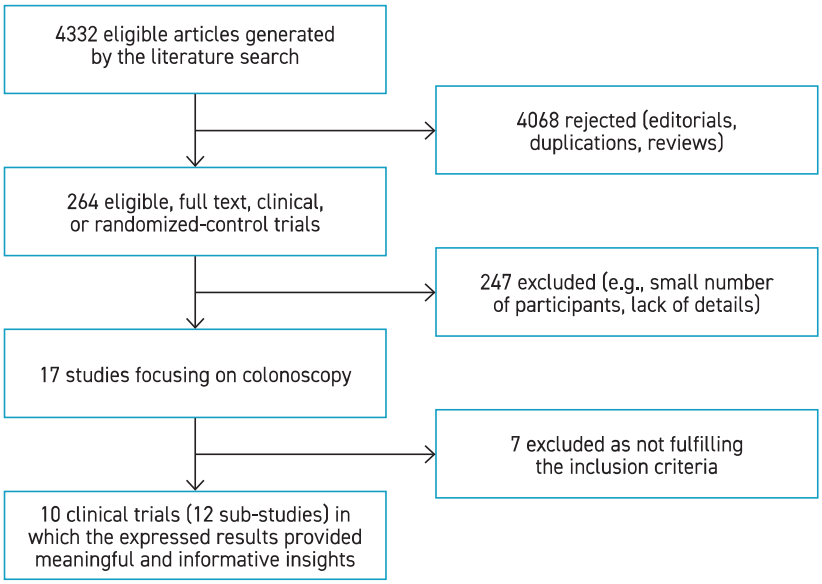


**Table 1.** Screening colonoscopy results

| Author | Sub-study | Number of patients | Average Risk Population | High Risk Population | Country | Time point | Men | Mean age, in years | Complete colonoscopy | ADR | SADR | AADR | CRC | Perforation rate | Bleeding rate |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Pox CP [15] | Six years of the German screening program | 2,821,392 | 2,821,392 | 0 | Germany | 2012 | 0.440 | 64.6 | 1.0000 | 0.194 | 0.0980 | 0.064 | 0.009 | 0.0002 | 0.0015 |
| Jover R [16] | Factors that influenced ADR in screening colonoscopies | 5059 | 5059 | 0 | Spain | 2013 | 0.436 | 59 | 0.9490 | 0.286 | 0.0000 | 0.114 | 0.000 | 0 | 0 |
| Butterly L [17] | New Hampshire Colonoscopy Registry, effect with withdrawal time | 7972 | 5521 | 2451 | USA | 2014 | 0.440 | 52 | 1.0000 | 0.336 | 0.0800 | 0.000 | 0.000 | 0 | 0 |
| Ross WA [18] | ADR and SSAR before and after implementation of quality improvement measures | 2833 | 2833 | 0 | USA | 2015 | 0.354 | 55 | 1.0000 | 0.410 | 0.0820 | 0.086 | 0.000 | 0 | 0 |
| Bretthauer M [19] | Participation rate, ADR, performance, and adverse events in European countries | 31,420 | 31,420 | 0 | Norway | 2016 | 0.500 | 60 | 0.9720 | 0.307 | 0.0230 | 0.104 | 0.005 | 0.0001 | 0.0014 |
| Zwink N [20] | Comparisons of colonoscopies in the German screening program | 5252 | 5252 | 0 | Germany | 2017 | 0.570 | 61 | 1.0000 | 0.300 | 0.0000 | 0.110 | 0.010 | 0.0006 | 0.0017 |
| Wong J [21] | ADR, AADR, APC in directs screening colonoscopy vs. FIT-positive colonoscopy | 2901 | 2901 | 0 | China | 2019 | 0.538 | 60.6 | 1.0000 | 0.375 | 0.0000 | 0.049 | 0.001 | 0 | 0 |
| Kim S [22] | Cap-assisted chromoendoscopy compared to standard colonoscopy | 949 | 923 | 26 | Korea | 2020 | 0.480 | 57.5 | 0.9920 | 0.449 | 0.0160 | 0.097 | 0.012 | 0.0000 | 0.0010 |
| Kim S [28] | Cap-assisted chromoendoscopy compared to standard colonoscopy | 935 | 913 | 22 | Korea | 2020 | 0.480 | 57.6 | 0.9860 | 0.544 | 0.0150 | 0.088 | 0.012 | 0.0000 | 0.0030 |
| Ladabaum U [23] | ADR extended to all screening/ surveillance score, and automated colonoscopy audit | 15,253 | 5117 | 0 | USA | 2021 | 0.464 | 53 | 0.9770 | 0.363 | 0.0720 | 0.067 | 0.000 | 0 | 0 |
| Xu H [24] | Artificial intelligence assisted colonoscopy compared to conventional colonoscopy | 1540 | 1540 | 0 | China | 2023 | 0.473 | 57.03 | 0.9870 | 0.335 | 0.0150 | 0.051 | 0.006 | 0 | 0 |
| Xu H [24] | Artificial intelligence assisted colonoscopy compared to conventional colonoscopy | 1519 | 1519 | 0 | China | 2023 | 0.465 | 57.49 | 0.9870 | 0.397 | 0.0130 | 0.067 | 0.002 | 0 | 0 |

ADR = adenoma detection rate, AADR = advanced adenoma detection rate, APC = adenomas per colonoscopy, CRC = colorectal cancer, FIT = fecal immunochemical, SADR = sessile adenoma detection rate

Thus, at least one adenoma was found in 35.3% of the patients, and CRC in 0.5%. ER for ADR was well inside the range of distribution of true effects [Figure 2].

The Q-statistic provides a test of the null hypothesis that all studies in the analysis share a common effect size. If all studies shared the same true effect size, the expected value of Q would be equal to the degrees of freedom (the number of studies minus 1). The Q-value was 7285.261 with 11 degrees of freedom and $P < 0.001$. Using a criterion alpha of 0.100, we can reject the null hypothesis that the true effect size is the same in all these studies. The $I^2$ statistic was 100%, which tells us that some 100% of the variance in observed effects reflects variance in true effects rather than sampling error. Tau-square, the variance of true effect sizes, was 0.269 in logit units. Tau, the standard deviation of true effect sizes, was 0.519 in

**Table 2.** Event rates, 95% confidence intervals, and forest plots comparing screening colonoscopies

| Study name and reference number | Subgroup within the study | Comparison | Time point | Event rate | 95% confidence interval | Forest plot |
|---|---|---|---|---|---|---|
| **Adenoma detection rate** | | | | | | |
| Pox CP [15] | Six years of the German screening program | Germany | 2012 | 0.194 | 0.194–0.194 | |
| Jover R [16] | Factors that influenced ADR in screening colonoscopies | Spain | 2013 | 0.286 | 0.274–0.299 | |
| Butterly L [17] | New Hampshire Colonoscopy Registry, effect with withdrawal time | USA | 2014 | 0.336 | 0.324–0.349 | |
| Ross WA [18] | ADR and SSAR before and after implementation of quality improvement measures | USA | 2015 | 0.410 | 0.392–0.428 | |
| Bretthauer M [19] | Participation rate, ADR, performance, and adverse events in European countries | Norway | 2016 | 0.307 | 0.302–0.312 | |
| Zwink N [20] | Comparisons of colonoscopies in the German screening program | Germany | 2017 | 0.300 | 0.288–0.313 | |
| Wong J [21] | ADR, AADR, APC in directs screening colonoscopy vs. FIT-positive colonoscopy | China | 2019 | 0.375 | 0.358–0.393 | |
| Kim S [22] | Cap-assisted chromoendoscopy compared to standard colonoscopy | Korea | 2020 | 0.449 | 0.417–0.481 | |
| Kim S [28] | Cap-assisted chromoendoscopy compared to standard colonoscopy | Korea | 2020 | 0.544 | 0.512–0.576 | |
| Ladabaum U [23] | ADR extended to all screening/surveillance score, and automated colonoscopy audit | USA | 2021 | 0.363 | 0.350–0.376 | |
| Xu H [24] | Artificial intelligence assisted colonoscopy compared to conventional colonoscopy | China | 2023 | 0.335 | 0.312–0.359 | |
| Xu H [24] | Artificial intelligence assisted colonoscopy compared to conventional colonoscopy | China | 2023 | 0.397 | 0.289–0.422 | |
| Pooled | | | | 0.353 | 0.289–0.423 | |
| Prediction interval | | | | 0.353 | 0.141–0.645 | |
| **Advanced adenoma detection rate** | | | | | | |
| Pox CP [15] | Six years of the German screening program | Germany | 2012 | 0.064 | 0.064–0.064 | |
| Jover R [16] | Factors that influenced ADR in screening colonoscopies | Spain | 2013 | 0.114 | 0.106–0.123 | |
| Butterly L [17] | New Hampshire Colonoscopy Registry, effect with withdrawal time | USA | 2014 | 0.000 | 0.000–0.001 | |
| Ross WA [18] | ADR and SSAR before and after implementation of quality improvement measures | USA | 2015 | 0.086 | 0.076–0.097 | |
| Bretthauer M [19] | Participation rate, ADR, performance, and adverse events in European countries | Norway | 2016 | 0.104 | 0.101–0.107 | |
| Zwink N [20] | Comparisons of colonoscopies in the German screening program | Germany | 2017 | 0.110 | 0.102–0.119 | |
| Wong J [21] | ADR, AADR, APC in directs screening colonoscopy vs. FIT-positive colonoscopy | China | 2019 | 0.049 | 0.042–0.057 | |
| Kim S [22] | Cap-assisted chromoendoscopy compared to standard colonoscopy | Korea | 2020 | 0.098 | 0.080–0.118 | |
| Kim S [28] | Cap-assisted chromoendoscopy compared to standard colonoscopy | Korea | 2020 | 0.088 | 0.071–0.108 | |
| Ladabaum U [23] | ADR extended to all screening/surveillance score, and automated colonoscopy audit | USA | 2021 | 0.067 | 0.060–0.074 | |
| Xu H [24] | Artificial intelligence assisted colonoscopy compared to conventional colonoscopy | China | 2023 | 0.051 | 0.041–0.064 | |
| Xu H [24] | Artificial intelligence assisted colonoscopy compared to conventional colonoscopy | China | 2023 | 0.067 | 0.056–0.081 | |
| Pooled | | | | 0.076 | 0.062–0.093 | |
| Prediction interval | | | | 0.076 | 0.035–0.160 | |

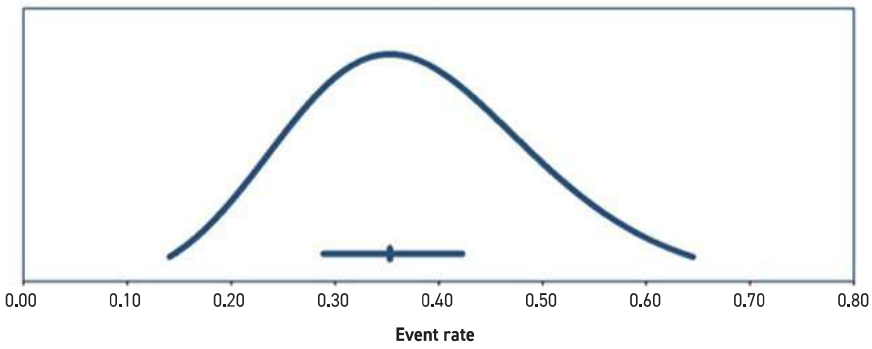
| Study name and reference number | Subgroup within the study | Comparison | Time point | Event rate | 95% confidence interval | Forest plot |
|---|---|---|---|---|---|---|
| **Sessile serrated adenoma detection rate** | | | | | | |
| Pox CP [15] | Six years of the German screening program | Germany | 2012 | 0.098 | 0.098–0.098 | |
| Jover R [16] | Factors that influenced ADR in screening colonoscopies | Spain | 2013 | 0.000 | 0.000–0.002 | |
| Butterly L [17] | New Hampshire Colonoscopy Registry, effect with withdrawal time | USA | 2014 | 0.080 | 0.073–0.088 | |
| Ross WA [18] | ADR and SSAR before and after implementation of quality improvement measures | USA | 2015 | 0.082 | 0.072–0.093 | |
| Bretthauer M [19] | Participation rate, ADR, performance, and adverse events in European countries | Norway | 2016 | 0.023 | 0.021–0.025 | |
| Zwink N [20] | Comparisons of colonoscopies in the German screening program | Germany | 2017 | 0.000 | 0.000–0.002 | |
| Wong J [21] | ADR, AADR, APC in directs screening colonoscopy vs. FIT-positive colonoscopy | China | 2019 | 0.000 | 0.000–0.003 | |
| Kim S [22] | Cap-assisted chromoendoscopy compared to standard colonoscopy | Korea | 2020 | 0.016 | 0.010–0.027 | |
| Kim S [28] | Cap-assisted chromoendoscopy compared to standard colonoscopy | Korea | 2020 | 0.015 | 0.009–0.026 | |
| Ladabaum U [23] | ADR extended to all screening/surveillance score, and automated colonoscopy audit | USA | 2021 | 0.072 | 0.065–0.079 | |
| Xu H [24] | Artificial intelligence assisted colonoscopy compared to conventional colonoscopy | China | 2023 | 0.015 | 0.010–0.022 | |
| Xu H [24] | Artificial intelligence assisted colonoscopy compared to conventional colonoscopy | China | 2023 | 0.013 | 0.009–0.020 | |
| Pooled | | | | 0.023 | 0.014–0.035 | |
| Prediction interval | | | | 0.023 | 0.004–0.126 | |
| **Colorectal cancer** | | | | | | |
| Pox CP [15] | Six years of the German screening program | Germany | 2012 | 0.009 | 0.009–0.009 | |
| Jover R [16] | Factors that influenced ADR in screening colonoscopies | Spain | 2013 | 0.000 | 0.000–0.002 | |
| Butterly L [17] | New Hampshire Colonoscopy Registry, effect with withdrawal time | USA | 2014 | 0.000 | 0.000–0.001 | |
| Ross WA [18] | ADR and SSAR before and after implementation of quality improvement measures | USA | 2015 | 0.000 | 0.000–0.003 | |
| Bretthauer M [19] | Participation rate, ADR, performance, and adverse events in European countries | Norway | 2016 | 0.005 | 0.004–0.006 | |
| Zwink N [20] | Comparisons of colonoscopies in the German screening program | Germany | 2017 | 0.010 | 0.008–0.013 | |
| Wong J [21] | ADR, AADR, APC in directs screening colonoscopy vs. FIT-positive colonoscopy | China | 2019 | 0.001 | 0.000–0.003 | |
| Kim S [22] | Cap-assisted chromoendoscopy compared to standard colonoscopy | Korea | 2020 | 0.012 | 0.007–0.021 | |
| Kim S [28] | Cap-assisted chromoendoscopy compared to standard colonoscopy | Korea | 2020 | 0.012 | 0.007–0.022 | |
| Ladabaum U [23] | ADR extended to all screening/surveillance score, and automated colonoscopy audit | USA | 2021 | 0.000 | 0.000–0.002 | |
| Xu H [24] | Artificial intelligence assisted colonoscopy compared to conventional colonoscopy | China | 2023 | 0.006 | 0.003–0.011 | |
| Xu H [24] | Artificial intelligence assisted colonoscopy compared to conventional colonoscopy | China | 2023 | 0.002 | 0.001–0.006 | |
| Pooled | | | | 0.005 | 0.004–0.007 | |
| Prediction interval | | | | 0.005 | 0.002–0.015 | |

ADR = adenoma detection rate, AADR = advanced adenoma detection rate, APC = adenomas per colonoscopy, CRC = colorectal cancer, FIT = fecal immunochemical, SADR = sessile adenoma detection rate

logit units. If the true effects are normally distributed (in logit units), we estimated that the prediction interval was 0.141–0.645. The true effect size in 95% of all comparable populations falls in this interval.

We measured sensitivity by excluding individual studies and recalculating the overall meta-analysis outcome. This process was repeated for each of the studies. Deviations from the primary result were not significant with ERs varying between 0.194 95%CI 0.194–0.194 and 0.544 95%CI 0.512–0.576, compared with the primary score of ER at 0.353, 95%CI 0.289–0.423. Calculating separately studies with low ERs, 0.194–0.336, and those with high ERs,

**Figure 2.** Distribution of true effect in adenoma detection rate
Mean effect size: 0.35 with a 95% confidence interval of 0.29–0.42
True effect size in 95% of all comparable populations: 0.14–0.65




0.336–0.544, we found ER 0.290, 95%CI 0.224–0.367, and ER 0.421, 95%CI 0.379–0.464, respectively. The prediction value was 0.099 to 0.602, thus both ERs were in this range. When the article by Pox and co-authors [15] was excluded from the meta-analysis, ER was 0.369, 95%CI 0.338–0.402, with a prediction interval of 0.254–0.502, well in the range of the meta-analysis included in this study. In addition, we measured the quality of the studies using the MINORS method. Scores between 1.63 and 1.92 were found with a median of 1.75. We separately performed a meta-analysis of 6 sub-studies with scores of 1.63 to 1.75, and 6 sub-studies with scores of 1.75 to 1.92. ORs were 0.325, 95%CI 0.245–0.416, and 0.382, 95%CI 0.325–0.442, respectively. Colonoscopy was complete and reached the cecum in 97% to 99% of the cases. Perforation and bleeding occurred in 627 cases (0.022%) and 4289 (0.148%) cases, respectively.

**COLONOSCOPY REMAINS A FEASIBLE AND SUITABLE SCREENING TOOL FOR ASYMPTOMATIC POPULATIONS.**

## DISCUSSION

Screening methods for CRC have been proven to be very beneficial in detecting pre-cancerous precursors and early-stage cancer, thereby improving the lifespan of patients [25]. Colonoscopy is considered a highly efficient method for detecting and removing pre-cancerous adenomas.

In this meta-analysis, we investigated the results of various studies focusing on screening colonoscopy for asymptomatic populations over the last 15 years, evaluating both efficacy and safety and comparing the findings with data from our previous meta-analysis conducted in 2008 [10]. Our analysis revealed a significant and noteworthy increase in the diagnostic yield of CRC precursors. We noticed a remarkable improvement in ADR by over 80% (0.353 compared to 0.190) and AADR (0.076 compared to 0.050). Several factors could have contributed to this observed improvement in detection rates including the enhancement of training methods and the implementation of computer-guided training [26] and improved equipment, particularly the introduction of a cap [22]. Furthermore, the integration of cutting-edge technologies such as artificial intelligence (AI) has marked a significant stride forward [24,27].

We found a substantial decline in the detection rate of CRC over the past 15 years, from 0.0078 to 0.005. This observation may be linked to the considerable increase in ADR, SSADR, and AADR, coupled with the subsequent removal of pre-malignant precursors that could potentially evolve into CRC. Although improvements were established in the field of polypectomy techniques, bleeding as a complication has been shown to increase over the years, rising from 0.05% to 0.15%. This increase could be attributed to improvements in detection rates and an associated increase in overall polypectomy rates. The increased awareness of postoperative complications could also contribute to this rise in numbers [28]. Another possible explanation is the increased usage of new oral anticoagulants in recent years [29]. Perforation rates have remained unchanged despite the potentially increased rate of polypectomy. This finding may be influenced by a combination of factors. One key factor could be the advancement in perforation preventive methods, such as the use of endoscopic closure devices like through-the-scope clips [30].

The major limitation of our review study is the challenge of predicting the percentage of usage of advanced techniques and measures, such as cap or AI, because only two representative articles about these methods were avail-

able for analysis. Another weakness is the small proportion of individuals with higher-than-average risk included in the statistics. A separate calculation for these two groups could not be performed as the data were uniformly presented for both. This review has several strengths, first, it was conducted following an a priori protocol, adhering to the PRISMA guidelines for a structured and transparent reporting process. Second, despite differences in demographics, which hold significant global representation, the cohorts in the chosen studies displayed similarities. This comparison included a balanced male-to-female ratio of 1:1, contributing to a highly realistic representation. We analyzed outcomes of screening colonoscopy in an asymptomatic population over 15 years, focusing on detection rates and complications.

## CONCLUSIONS

Colonoscopy screening for asymptomatic individuals demonstrates enhanced efficacy in both detecting and preventing CRC. Although colonoscopy has significantly improved its efficacy, it remains associated with procedural complications, such as the observed increase in bleeding. Colonoscopy remains a feasible and suitable screening tool for asymptomatic populations.

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